

CC2652R7 Technical Documentation RAG System

Corpus: Texas Instruments CC2652R7 wireless MCU documentation **System:** local retrieval-augmented generation pipeline, Python 3.12, Ollama Llama 3.2 3B fallback **Date:** May 2026

1. Corpus and Task

The corpus is public Texas Instruments documentation for the CC2652R7 SimpleLink wireless microcontroller. It was chosen because device-specific firmware and hardware questions are easy for a general LLM to confuse with nearby CC26xx variants, while the official documents contain exact values, register details, and SDK constraints. The indexed data contains three documents and 2361 processed chunks:

Source	Role	Chunks
data/raw/cc2652r7.pdf	CC2652R7 datasheet	60
data/raw/swcu192.pdf	CC13x2x7/CC26x2x7 TRM	2237
data/raw/Users_Guide.html	SimpleLink SDK User's Guide	64
Total		2361

The system is intended to answer firmware-debugging, hardware-specification, register-level, API-usage, absence, and comparison questions. The most important single chunk is `datasheet_hier_chunk_0000`, which contains the datasheet feature list: 48-MHz Arm Cortex-M4F, 704 KB flash, 144 KB SRAM, Bluetooth 5.2 Low Energy, IEEE 802.15.4/Zigbee/Thread/Matter support, 1.8-V to 3.8-V supply, and 31 GPIOs. Important corpus gaps remain: the TI RF Driver API Reference is not indexed, and competitor datasheets such as CC2652R1/CC2652P/CC1352R are not indexed.

2. System Architecture

The implementation exposes `RAGSystem.answer(question)` and returns `answer`, `sources`, `retrieved_chunks`, plus diagnostic trace and validation fields. The pipeline is manual rather than a black-box framework wrapper:

Stage	Component	Purpose
1	Dense retrieval	FAISS search over normalized BGE embeddings, candidate $k=20$
2	BM25 retrieval	Keyword retrieval with <code>BM25Okapi</code> , candidate $k=20$
3	Hybrid merge	Deduplicate by <code>chunk_id</code> , keep the max score, top 20
4	Identifier detection	Detect firmware symbols, hex addresses, register-like names, and all-caps identifiers
5	Reranking	Cross-encoder rerank with exact identifier-match boost
6	Anchor and budget	Prepend datasheet feature chunk for spec questions, deduplicate, keep about 2000 words
7	Generation	Remove table-of-contents chunks, run deterministic extractors, then use local Llama fallback

The anchor step is intentionally after reranking. Earlier experiments showed that injecting the broad datasheet feature list before the cross-encoder could demote it for focused peripheral queries; prepending it after reranking ensures high-value feature facts survive the context budget.

3. Chunking

Two chunking strategies were implemented. The baseline fixed strategy uses 512-word chunks with 64-word overlap. It is simple and reproducible, but ignores document structure and can split tables or sections awkwardly.

The current indexed strategy is hierarchical and section-aware. PDF pages and HTML h2/h3 sections are kept intact when they are at most 600 words; longer units are split into approximately 500-word subchunks. The code uses word counts, although some planning notes use the word "tokens." Each chunk stores `chunk_id`, `doc_id`, `text`, and metadata such as source file, page, section, and chunking strategy.

Chunking helped because the datasheet first-page feature list remains a compact chunk, allowing many spec answers to be cited from `datasheet_hier_chunk_0000`. It hurt in table-heavy cases: the "maximum RF output power" question currently retrieves a transmit-current table cell and answers `+0 dBm`, while the expected answer is `+20 dBm`. No chunking strategy can fix the RF Driver API failures because that source document is absent.

4. Embedding, Indexing, and Retrieval

Embeddings are produced with `BAAI/bge-large-en-v1.5`, normalized, and stored in a FAISS `IndexFlatIP`. Inner product over normalized vectors is equivalent to cosine similarity and is deterministic enough for a local reproducible index. A BM25 index is also built over the same chunks and stored with the processed data. `src/build_index.py` rebuilds the processed artifacts from the raw corpus.

Retrieval combines semantic and lexical evidence. Dense retrieval handles paraphrases such as "supply range" versus "single supply voltage"; BM25 preserves exact symbols and numeric strings. The merged candidate set is reranked by `cross-encoder/ms-marco-MiniLM-L-6-v2`. If the query contains identifiers such as `RF_open`, `RFCCpePatchExp`, register names, or hex values, chunks with exact identifier matches receive an additional +3.0 boost. This is important for embedded documentation, where exact symbol identity matters more than general semantic similarity.

5. Prompt and Generation Design

Generation is extractor-first. Before calling the LLM, the system removes table-of-contents chunks and tries deterministic extractors for common device facts: memory sizes, protocols, unsupported connectivity, CPU, clock, voltage, package, temperature, RF core, BLE sensitivity, GPIO count, ADC, serial interfaces, timers, TX power, and RF command chaining. These extractors return a strict cited format:

```
QUOTE: ...
ANSWER: ...
SOURCE: chunk_id
```

If no extractor applies, the local Ollama llama3.2 model is called with temperature=0, top_p=0.1, and num_predict=180. The prompt requires answers to use only retrieved context, cite chunks, avoid guesses, avoid Wi-Fi/Wi-SUN conflation, and say that information was not found when evidence is missing. A validation pass checks that technical literals such as units, hex values, and symbols in the answer appear in retrieved context.

6. Evaluation Results

The gold set contains 50 questions: 10 factual, 10 numerical, 10 negation/absence, 10 comparison, and 10 debugging. The latest saved evaluation uses k=5.

Metric	Value	Count	Caveat
Hit@5	1.000	50/50	Not discriminative: all 50 gold entries have empty must_cite_chunk_ids
Answerable@Context	0.560	28/50	Checks reference key terms in context, not full answer correctness

Category	N	Hit@5	Answerable@Context
Numerical	10	1.000	0.900
Factual	10	1.000	0.700
Negation	10	1.000	0.600
Debugging	10	1.000	0.400
Comparison	10	1.000	0.200

These metrics must be interpreted carefully. Hit@5 currently proves that the evaluation harness ran, not that the retriever found labeled relevant chunks. Answerable@Context now normalizes spaces and hyphens for key terms, fixing the voltage false negative where references contain 1.8V and 3.8V while the corpus writes 1.8-V and 3.8-V; it still checks key-term presence rather than full answer correctness. Some reference answers are also wrong for CC2652R7: the gold set says 256 KB SRAM, but the indexed datasheet says 144 KB; it also contains an 85 deg C temperature expectation while current corpus evidence supports -40 to +105 deg C.

Manual inspection of 12 latest answers found 6 correct, 2 partially correct, 3 incorrect, and 1 unsupported/hallucinated. Correct answers include flash size, SRAM size, voltage range, package, UART count, and Wi-Fi absence. Failures include maximum RF output power (+0 dBm instead of +20 dBm), standard-mode TX power, RF API questions outside the corpus, and generic hard-fault debugging answers.

7. Ablation Study

Two ablation scripts were run for retrieval. Both completed after model access was available locally or via the approved environment.

Experiment	Retrieval/generation change	Hit@5	Answerable@Context	Interpretation
Full pipeline	Hybrid dense+BM25, rerank, anchor injection, extractor-first generation	1.000	0.560	Current headline run
Dense only	FAISS dense retrieval only	1.000	Not measured	Inconclusive because citation labels are empty
No rerank	Hybrid retrieval without cross-encoder reranking	1.000	Not measured	Inconclusive for the same reason

The ablation table satisfies the requirement to compare retrieval variants, but it is not yet a strong scientific result. A better next evaluation would label must_cite_chunk_ids for at least a focused subset and compute source-hit changes for dense-only, BM25-only, hybrid, and reranked retrieval.

8. Failure Analysis and Future Work

The failures split into three classes. First, some gold questions require sources that are not indexed. RF API questions about RF_open, RF_close, RFCCpePatchFxp, RF_EventLastCmdDone, and CPE patch order require the RF Driver API Reference. A grounded RAG system should refuse these until that source is added. Competitor comparisons have the same issue because CC2652R1, CC2652P, CC1352R, and CC2652RB datasheets are absent.

Second, some failures happen even though the answer is present. The standard-mode TX-power question is the clearest example: datasheet_hier_chunk_0001_sub0 says the CC2652R7 supports +5 dBm TX at 9.7 mA, and datasheet_hier_chunk_0030 maps TX power setting 5 to 4.8 dBm typical output power. The system instead retrieved TRM chunks about CMD_SET_TX20_POWER and the 20 dBm PA path, then said the answer was not directly stated. This does not prove the question is impossible for every RAG system; it shows that generic retrieval is not guaranteed to use the right evidence when terms such as "TX power", "PA", and "power" appear in several unrelated contexts.

Third, evaluation still hides root causes. Hit@5 is non-discriminative without source labels, Answerable@Context checks key-term presence rather than correctness, and some gold answers conflict with the indexed datasheet. Unsupported connectivity answers improved by anchoring the datasheet feature/protocol list for Wi-Fi, LTE/cellular, USB, Ethernet, and Bluetooth Classic, but absence claims remain reliable only when explicit corpus evidence supports them.

The improvement order should be: label required source chunks and report Source Hit@1, Source Hit@5, MRR, Precision@5, and Recall@5 on the labeled subset only; classify failures as missing-source, retrieval miss, generation error, or gold mismatch; add a TX-power-specific retrieval/extraction path that boosts dBm datasheet tables, penalizes contradictory 20 dBm PA evidence for "without PA" queries, normalizes 4.8 dBm and +5 dBm, and validates numeric answers against cited dBm evidence; add the RF Driver API Reference only with an index rebuild and full audit; add competitor datasheets or rewrite comparison questions; and strengthen strict-format validation for LLM fallback answers.